

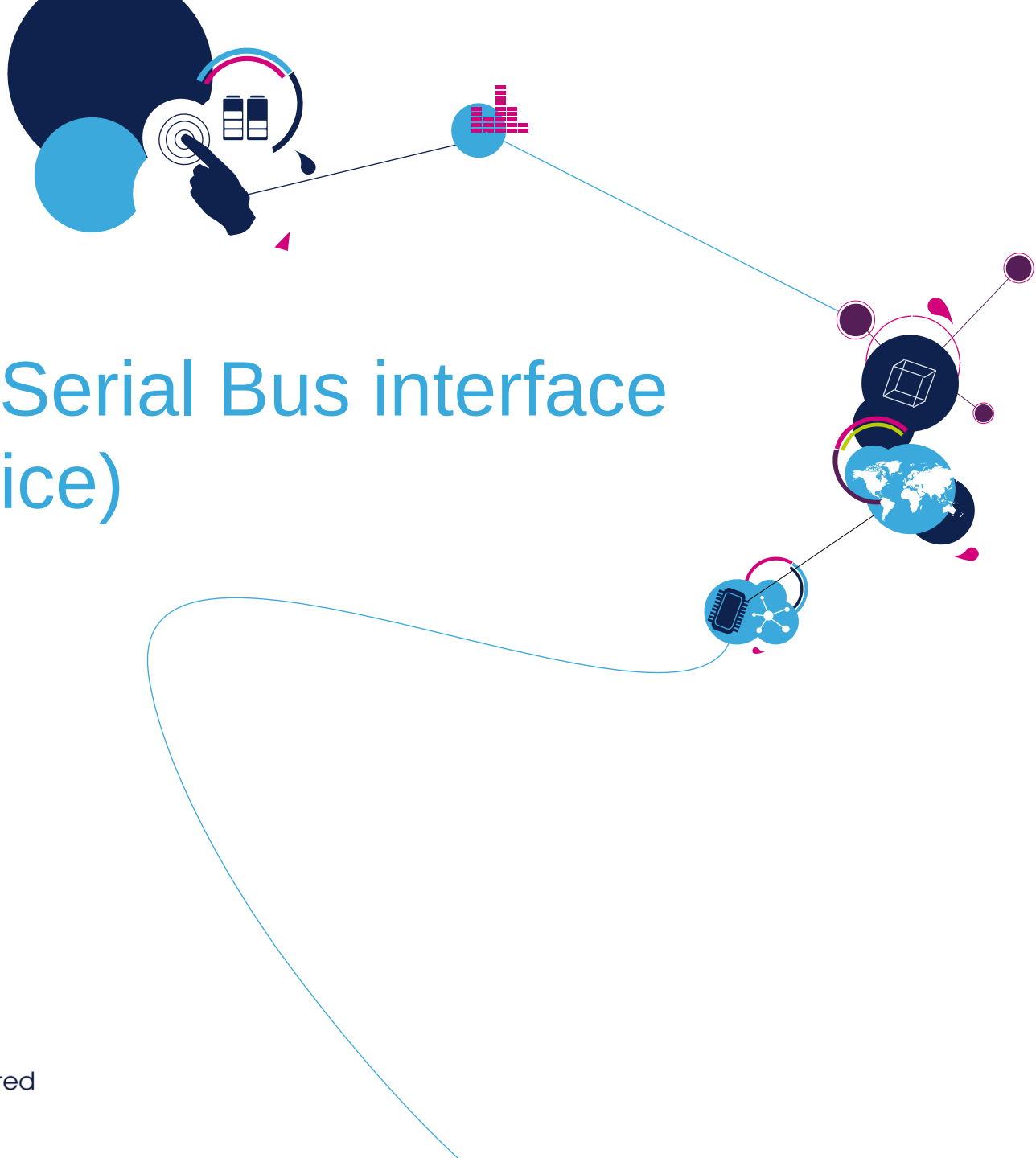


# STM32F3 Technical Training

**For reference only**

**Refer to the latest documents for details**





# Universal Serial Bus interface (USB Device)

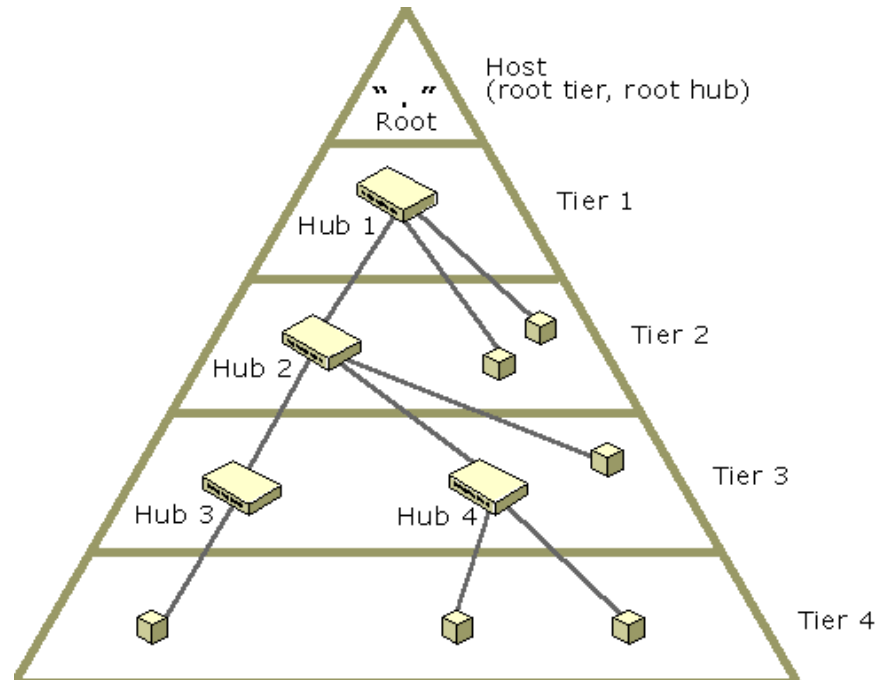
# USB Speeds & bus components

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- USB 2.0 speeds
  - Low speed: 1.5 Mbits/s
  - **Full speed: 12 Mbits/s**
  - High speed: 480 Mbits/s
- USB keeps high compatibility level between all supported speeds
- Bus components
  - **USB host or Root hub**: initiates all the transaction on the bus
  - **USB device**: is a set of one or more interfaces that expose capabilities to the host (ex: mouse, keyboard,..)
  - **USB hub**: allows to connect multiple devices to the USB host. It has an upstream port for communication with the host and multiple downstream ports for connection to devices
- Devices address assignment
  - After detecting a device attachment, the host will assign a unique address (on 7-bits) to the device

# USB 2.0 Bus Tiered-Star Topology

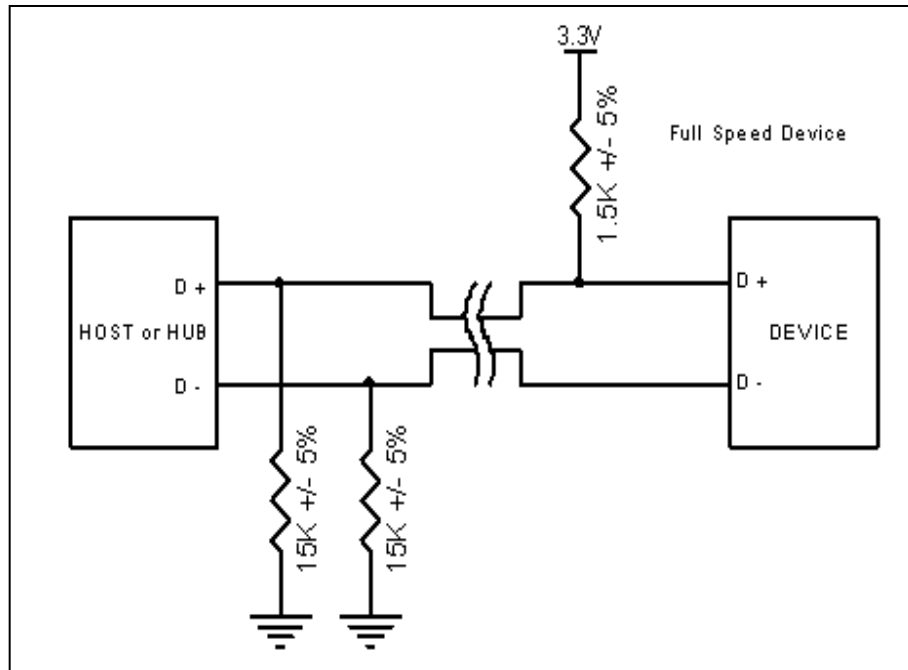
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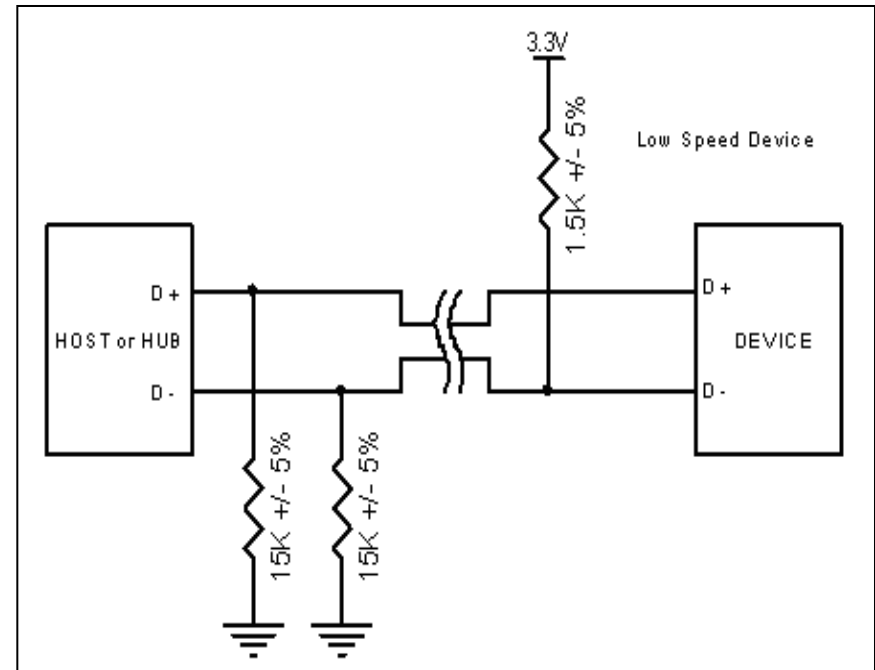
- A maximum of 127 devices can be connected in the bus
- A maximum of 5 hubs can be connected in series

# USB Device attachment & speed detection

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**Full/high Speed: Pull-up on D+**



**Low Speed: Pull-up on D-**

- The 1.5K pull-up allows the host to detect the device attachment and its supported speed
- High-speed device is detected first as full-speed device then high-speed capability is detected through bus handshake called “chirp sequence”

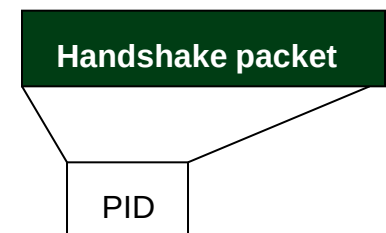
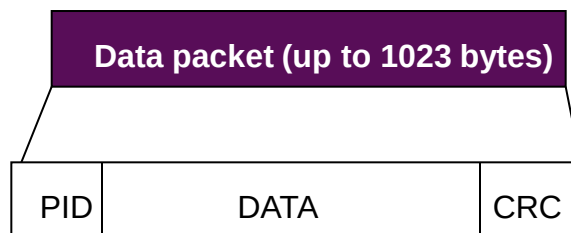
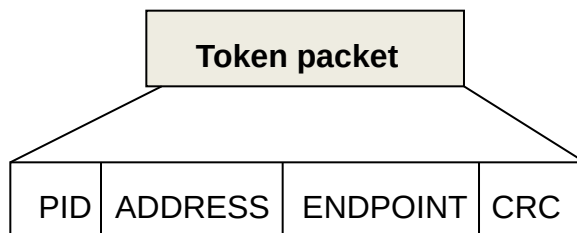
- Two possible power configurations
  - Self-powered: power provided from external power-supply
  - Bus-powered: power provided from VBUS (5v)
- For bus-powered device, two options are possible:
  - **Low-power devices** :maximum power consumption is 100mA
  - **High-power devices** :maximum power consumption is 100mA during bus enumeration and 500mA after configuration
- During device enumeration, the device indicates to host its power configuration (self-powered/bus-powered) and its power consumption in the device configuration descriptor

# USB Suspend mode

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- USB device should enter in USB Suspend mode when the bus is idle for more than 3 ms
- In suspend mode, the current drawn by the USB device from VBUS power shouldn't exceed 2.5mA ( old specification was 500μA)
- USB host prevents device from entering in suspend mode by periodically issuing Start of Frame (SOF) or Keep alive (for low-speed)
  - For High-speed, SOF is sent every micro-frame 125us +/- 65ns
  - For Full-speed, SOF is sent every frame 1ms +/- 500ns
  - For Low-speed, keep alive which is EOP (End of Packet) is sent every 1ms in absence of low-speed data
- Exist from Suspend mode can be
  - Initiated from host by issuing the resume signaling ( Resume or Reset)
  - Initiated from device by issuing the remote wakeup signaling

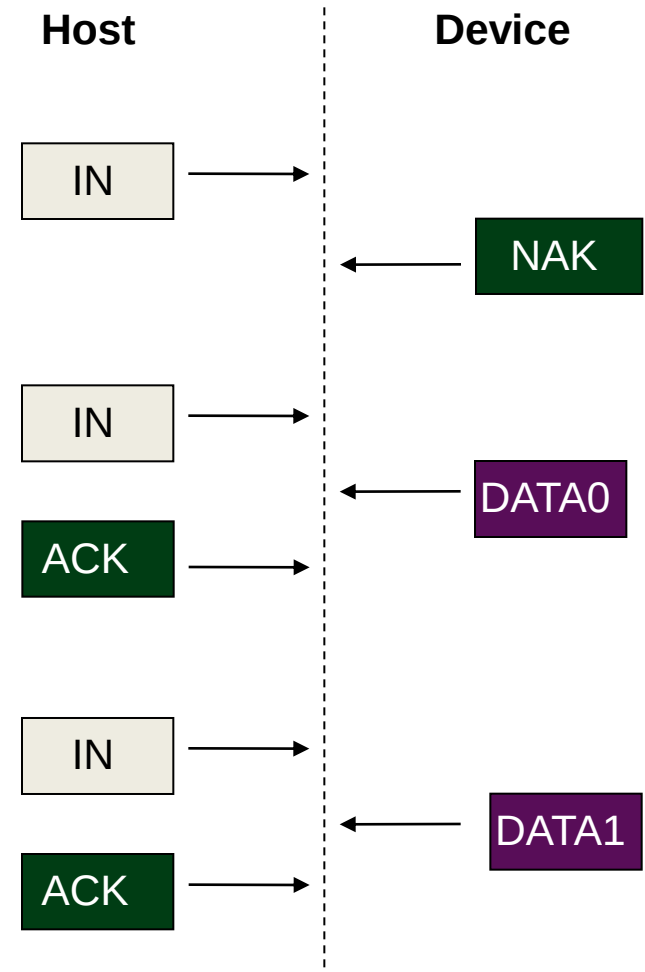
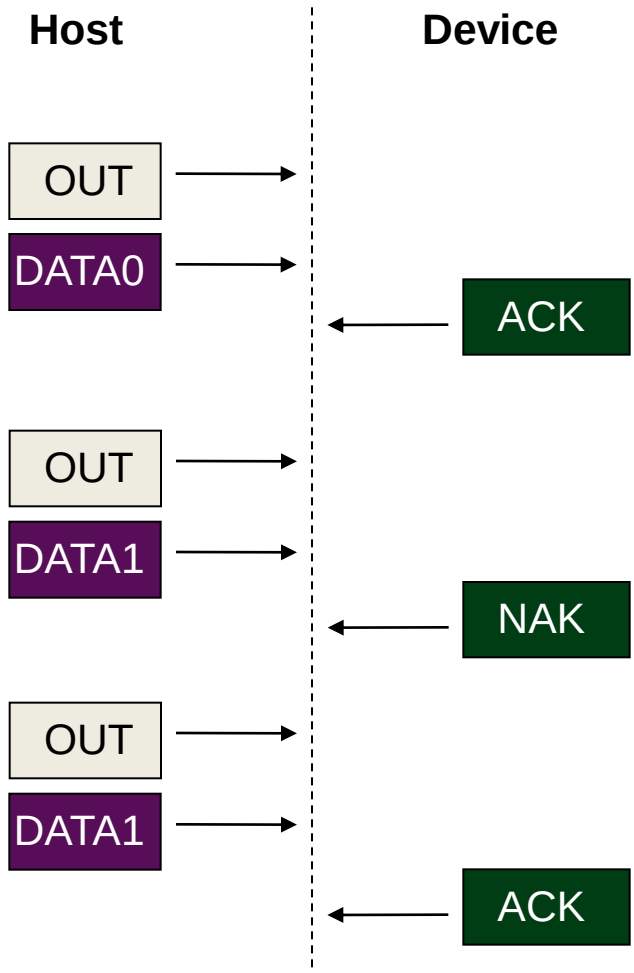
- One bus transaction is composed of a:
  - **Token packet (SETUP, IN, OUT)** always issued by the host
    - Target device address
    - Target endpoint number
    - Direction of transaction (IN: Device to host or SETUP/OUT: host to device)
  - **Data packet (DATA0, DATA1, DATA2, MDATA)**
    - Carries the data payload of a transaction sent by the host or device
    - DATA PID toggle used to synchronize HOST and DEVICE to avoid repeated packet transfer in case of corrupted/lost handshake
  - **Handshake packet (ACK, NAK, STALL, NYET)**
    - ACK: packet reception acknowledged (sent from host or device)
    - NAK : packet reception not acknowledged (sent from device)
    - STALL: control request not supported or endpoint halted (sent from device)
    - NYET: device not ready to accept further packets (only from high-speed device)





# Examples of IN/OUT transactions

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- A USB transfer is composed of one or multiple bus transactions
- Four types of USB transfers are defined:
  - **Control**: used for control and configuration requests (ex: device enumeration)
  - **Bulk**: used for huge data transfers with no guaranteed delivery rate (ex: printer, mass-storage drive,..)
  - **Interrupt**: used for interrupt driven devices that need to be polled periodically for small size data transfer (ex: mouse, keyboard, joystick)
  - **Isochronous**: used for data streaming applications, that requires a guaranteed delivery rate, but no error checking (ex: audio, video devices)
- During each frame (in LS/FS) or micro-frame (in HS), the host will schedule the needed transfers with different bandwidth allocation for each transfer type

# USB Control Transfer

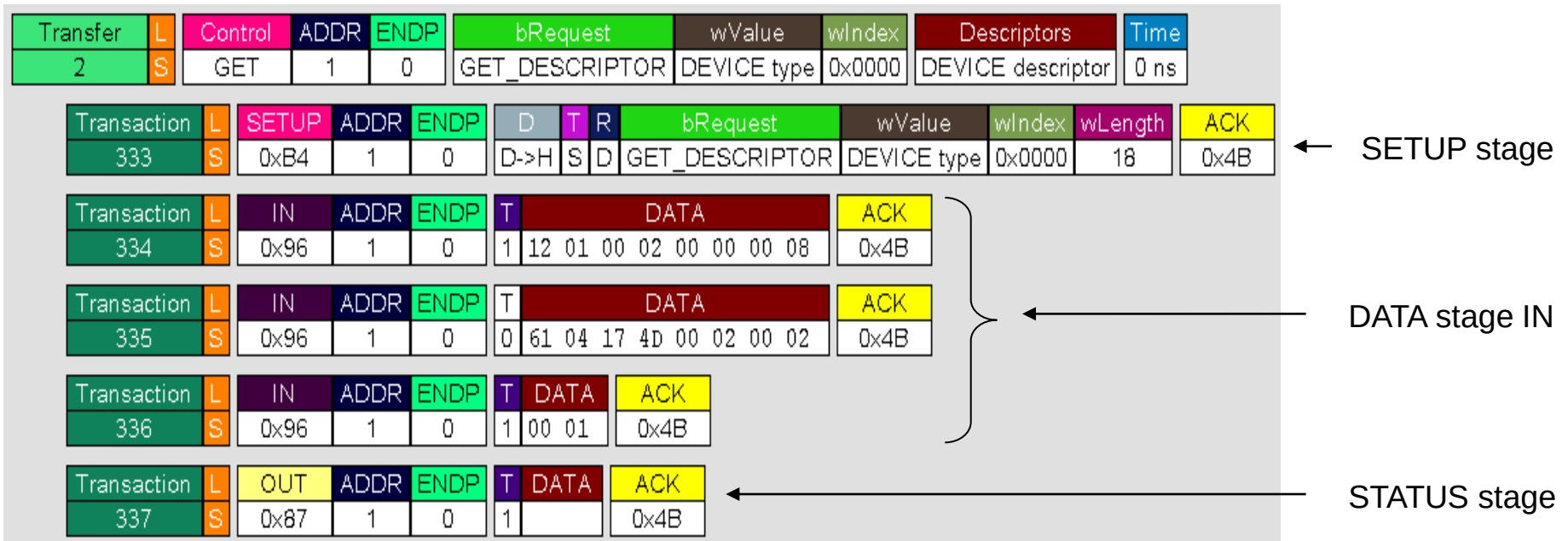
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- Used for standard control requests during device enumeration process or during class operation
- All devices should support control transfer through endpoint 0 (bidirectional)
- It is given reserved 10% of bus bandwidth for FS/LS and 20% for HS
- Control transfer has 3 stages
  - **SETUP stage:** one SETUP transaction for issuing the control request (ex: Get Descriptor)
  - **Optional DATA stage IN or OUT:** one or multiple data transactions
  - **Status stage:** one IN or OUT transaction with a Zero Length data packet to check if control transfer request executed correctly or not.
- **The maximum data packet size** during the optional data stage is 8 bytes for LS and **64 bytes in FS/HS**
- **Transfer error management** done through handshake packet and data PID toggle mechanism

# Example of a USB Control Transfer

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Get device descriptor standard request:



- Used to transfer large amount of data without guaranteed delivery rate (sending data to printer, drive,...)
- Lowest priority transfer with no reserved bus bandwidth but can occupy the full bandwidth if no other transfer on the bus
- Supported only by full-speed and high-speed devices
- Can consist of one or more IN or OUT transactions (one pipe/endpoint needed for each direction) during each frame/micro-frame
- **The max packet size is 64 bytes for FS and 512 bytes for HS**
- Used in most of the case as a transport layer for a higher application or class protocol layer (Bulk-Only transfer in mass-storage class)
- **Transfer error management** done through handshake packet and data PID toggle mechanism

# USB Interrupt Transfer

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- Useful when data need to be transferred with a maximum transfer latency (mouse, keyboard,...)
- IN or OUT data transfers can occur periodically within a maximum latency period negotiated during device enumeration
- Has a limited reserved bandwidth with a guaranteed maximum latency
  - For LS the packet max length is 8 bytes with a maximum latency of up to 1 packet each 10 frames
  - **For FS the packet max length is 64 bytes with a maximum latency of up to 1 packet each frame**
  - For HS the packet max length is 1023 bytes with a maximum latency up to 3 packets each micro-frame
- When bandwidth is available the host is free to schedule extra OUT or IN interrupt transactions (more than the predefined maximum latency period)
- **Transfer error management** done through handshake packet and data PID toggle mechanism

# USB Isochronous transfers

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- Used mainly for streaming **real-time data like audio and video**
- Needs a guaranteed transfer rate with a predefined bytes number in every frame/micro-frame but No transfer error checking
- The transfer rate is negotiated between host and device during enumeration
- Transfer can consist of one or more data OUT or IN transactions
  - **In Full-speed, the max packet length is 1023 bytes with a maximum of one packet per frame**
  - In High-speed, the max packet length is 1024 bytes with a maximums of 3 packets per frame
  - Isochronous transactions does not include a handshake packet
  - Not supported by low-speed devices
- Clock synchronization between the host and device may be needed (ex: audio speaker) it can be done by
  - Device synchronizing its clock to the SOF packet
  - Using a feedback pipe for flow control

# Interrupt & Isochronous Transfers Host Constraints

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- The host may not be able to provide the requested bandwidth to device, in this case the host will try other possible configurations with lower bandwidth requirements (if provided by the device)
- If still no bandwidth available, the host will refuse device configuration
- Host software may have some latency for processing data and issuing transfer requests on time due to other processes taking CPU time
- In order to avoid multiple SW calls for handling data to be transmitted or received, large chunks of data transfers should be scheduled



# USB controllers in the STM32 microcontroller series



# USB Device Controllers in STM32 series

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- USB device controller is present in almost all STM32 ARM cortex M3/M4 series
- Three hardware implementations are available
  - **USB 2.0 full-speed device controller**
  - USB 2.0 full-speed dual role host/device OTG controller
  - USB 2.0 high-speed dual role host/device OTG controller
- Selection of the controller that can the application will depends on
  - Needed USB transfer performance
  - Needed CPU performance
  - Available Flash and RAM memory size
  - Presence of other needed peripherals
  - Power consumption requirements
  - External components (BOM)



# USB Device Controller in STM32F1/F3/L1

**USB 2.0 Full-speed Device  
Controller**



# USB 2.0 Full-speed Device Controller Features

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- Available on the following ARM Cortex M3/M4 platforms:
  - **STM32F102: USB access line** (48 MHz MCU, up to 16KB SRAM and 128KB of FLASH )
  - **STM32F103: Performance line** (72 MHz MCU, up to 96KB SRAM and 1MB FLASH)
  - **STM32L152: Ultra-low power series** (32 MHz MCU, up to 16KB SRAM and 128KB of FLASH)
  - **STM32F3xx : DSP & Analog ( 72MHz MCU, up to 32KB of SRAM)**
- Main features
  - USB 2.0 full-speed compliant
  - Up to 8 bi-directional endpoints (or 16 unidirectional endpoints)
  - Embedded full-speed analog transceiver
  - Supports all transfer modes (control, bulk, interrupt and isochronous)
  - Dedicated SRAM area of 512 bytes as packet memory that can be shared among the needed endpoints
  - Double-buffering mechanism for isochronous and bulk transfers
  - USB Suspend/Resume with system entry/wakeup for low power mode

# USB 2.0 Full-speed Device Controller Block Diagram

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- **SIE (Serial Interface Engine)**

- NRZI Encoding/Decoding
- Synchronization & Pattern Recognition
- Bit-stuffing and Handshake evaluation
- PID & CRC generation and checking
- Interrupt generation

- **Suspend Timer**

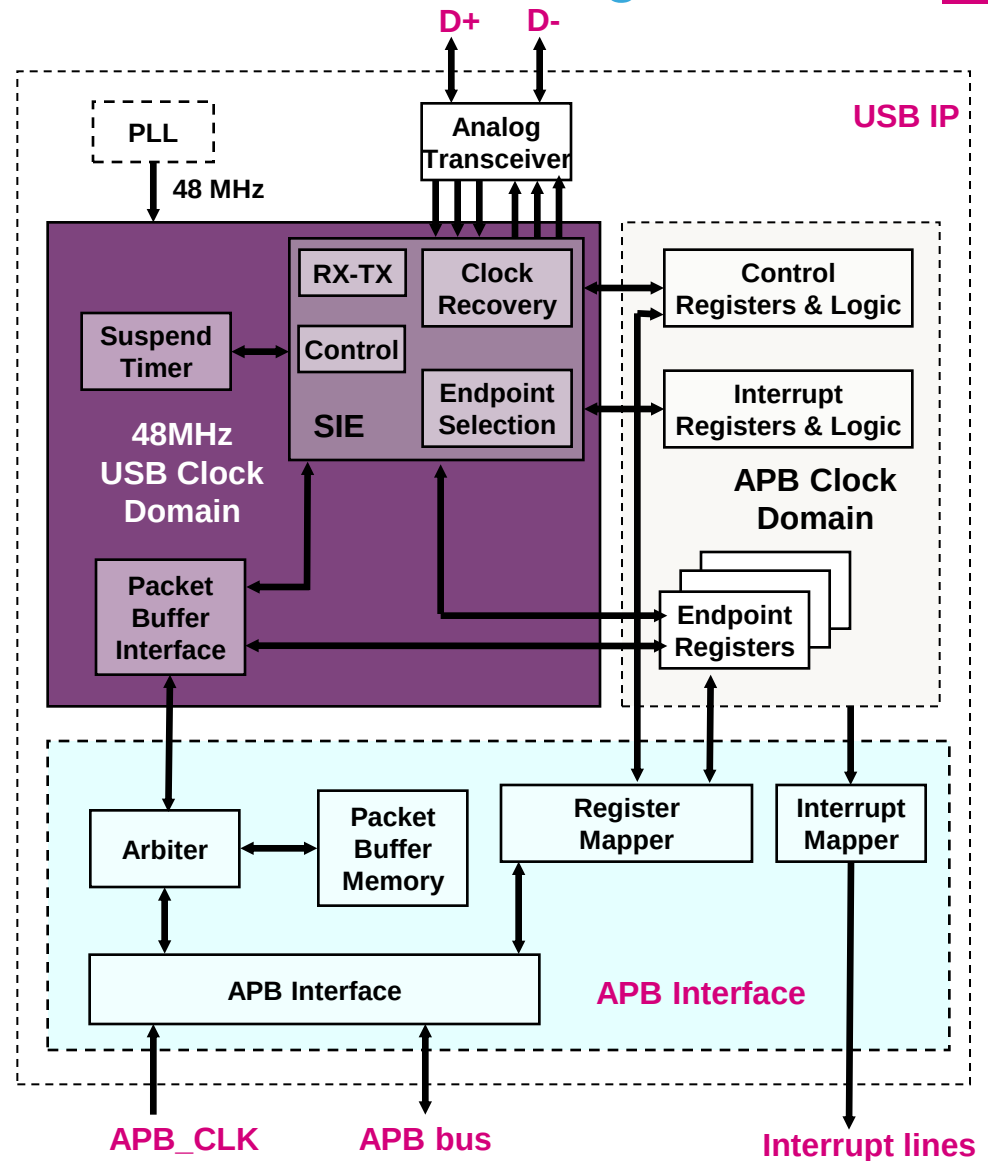
- Generate the Suspend interrupt when no SOF is detected for 3ms

- **Packet Buffer Memory**

- 512 bytes dedicated SRAM memory
- The Arbiter allows dual access either from packet buffer interface or APB interface

- **3 interrupt vectors (lines)**

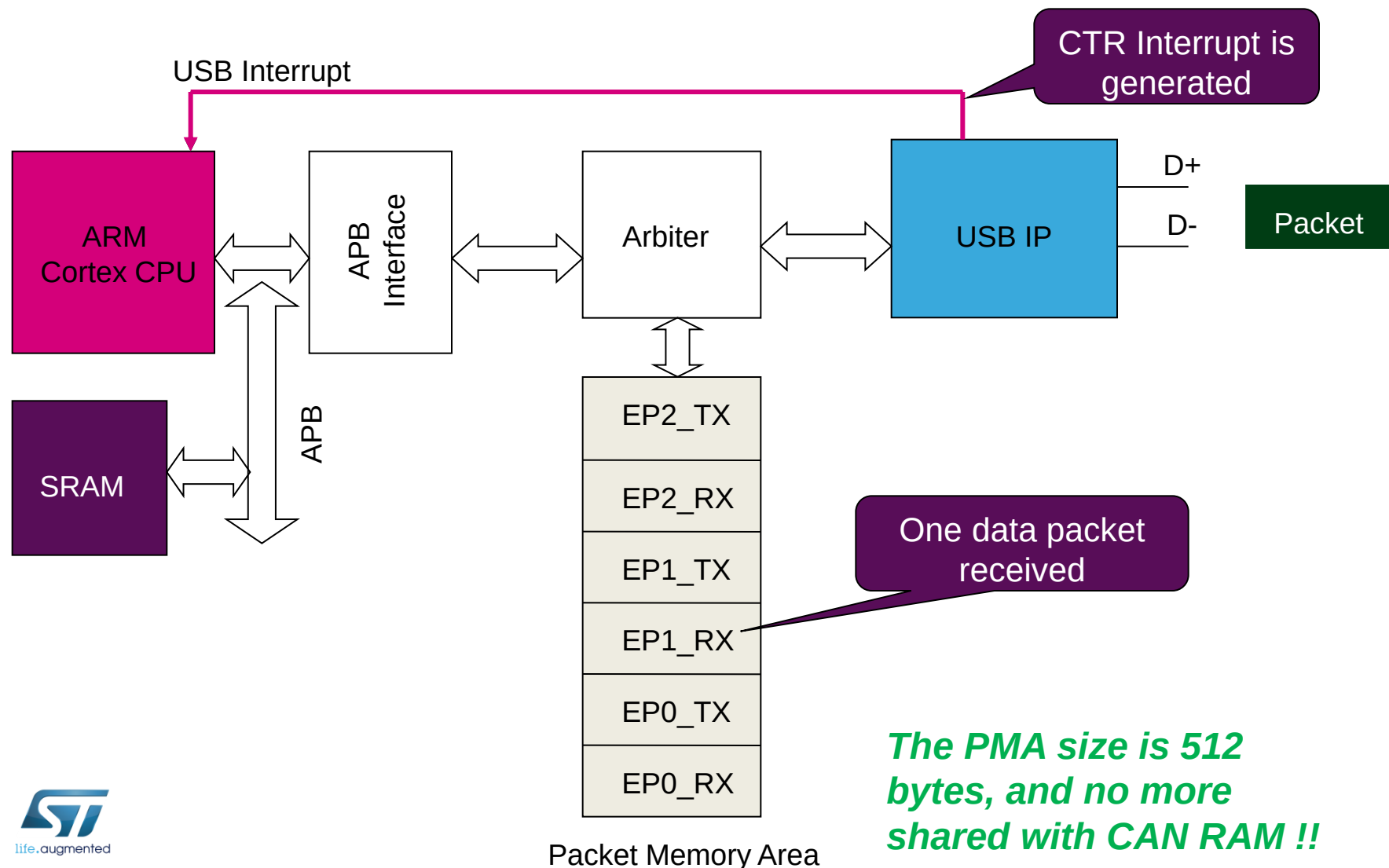
- Low priority interrupt for managing all endpoints
- High priority interrupt: can be used for managing isochronous/double-buffered endpoints only
- Suspend/Resume interrupt



# USB 2.0 Full-speed Device Controller

## Operation overview

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# USB 2.0 Full-speed Device Controller

## Transactional model handling

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- After each successful transaction on any configured endpoint, an interrupt (correct transfer CTR) is raised
- The “Correct transfer” interrupt handler has to:
  - Check interrupt status bits to determine the endpoint on which the transaction has occurred
  - For **OUT/SETUP endpoints**: copy received data packet from packet memory area to application buffer for processing, then re-enable the endpoint to be able to receive next incoming packet
  - For **IN endpoints**: copy next data to be transferred from application buffer to packet memory area, then re-enable the endpoint to send the packet when the next IN token comes from host
- The hardware will automatically change the endpoint to NAK state after end of each transaction, so it is up to application to re enable endpoint for next transaction
- The Transactional model has simple FW handling, but does not allow multiple-packet transfer without CPU intervention after each transferred packet

# USB 2.0 Full-speed Device Controller Endpoint Configuration/Enabling

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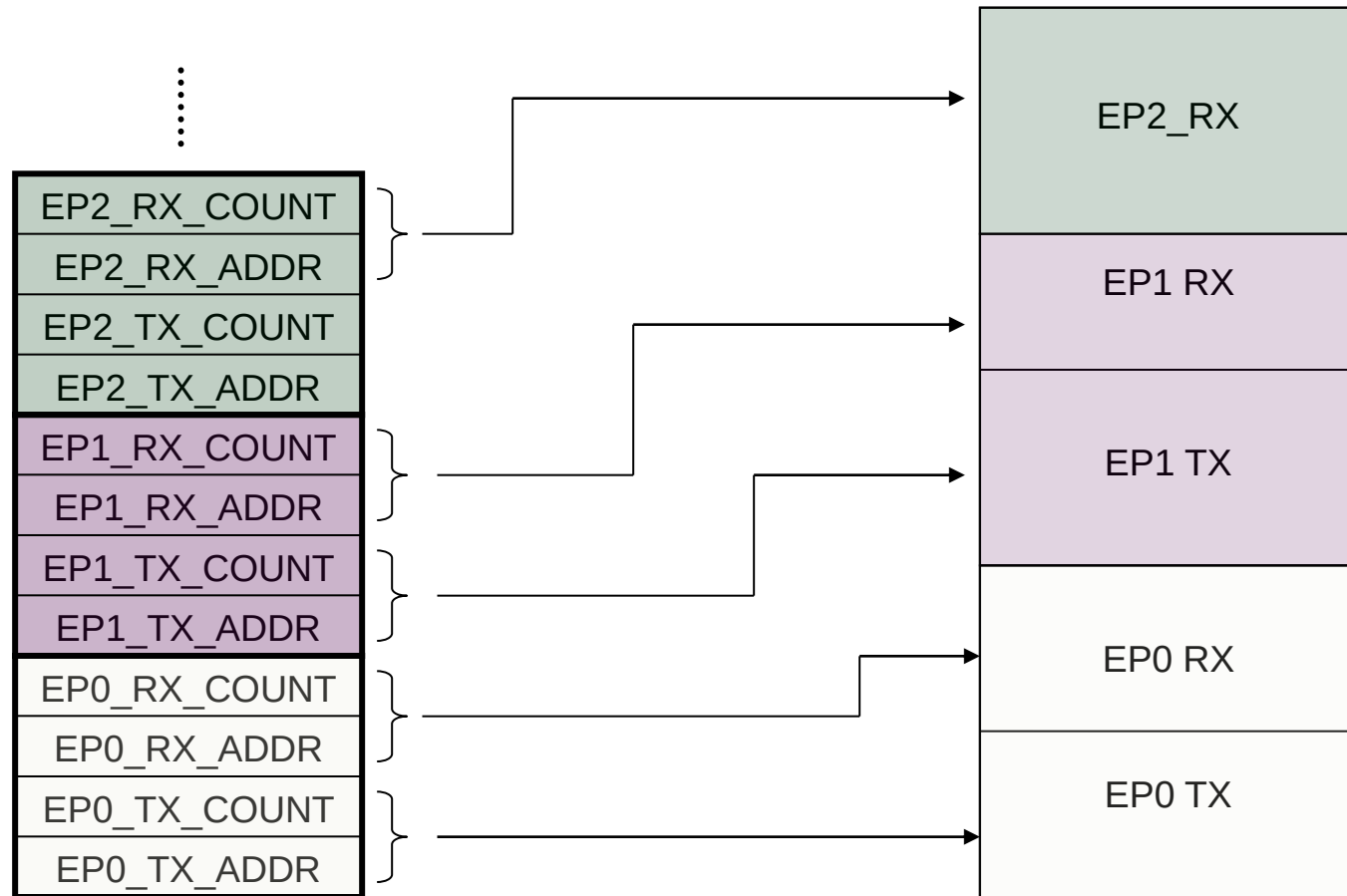
- Before start of any transfer on one endpoint, the following configuration should be done:
  - Endpoint address (only lower four bits)
  - Endpoint transfer type (control, bulk, interrupt or isochronous)
  - Endpoint TX or RX packet start address location in the packet memory area
  - For OUT/SETUP endpoints the max receive packet size should be configured
- After the configuration, endpoint can be enabled for a transfer
  - **IN endpoint:**
    - Data can be copied from application buffer to endpoint PMA buffer
    - the TX transfer count should be updated (the maximum is one max packet size)
    - Endpoint status should be changed to “ACK” to allow data transfer when IN token arrives
  - **OUT/SETUP endpoint:**
    - Endpoint status should be changed to “ACK” to allow OUT/SETUP data packet reception on endpoint



# USB 2.0 Full-speed Device Controller

## Packet Memory Area

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Buffer Description Table (BTABLE)

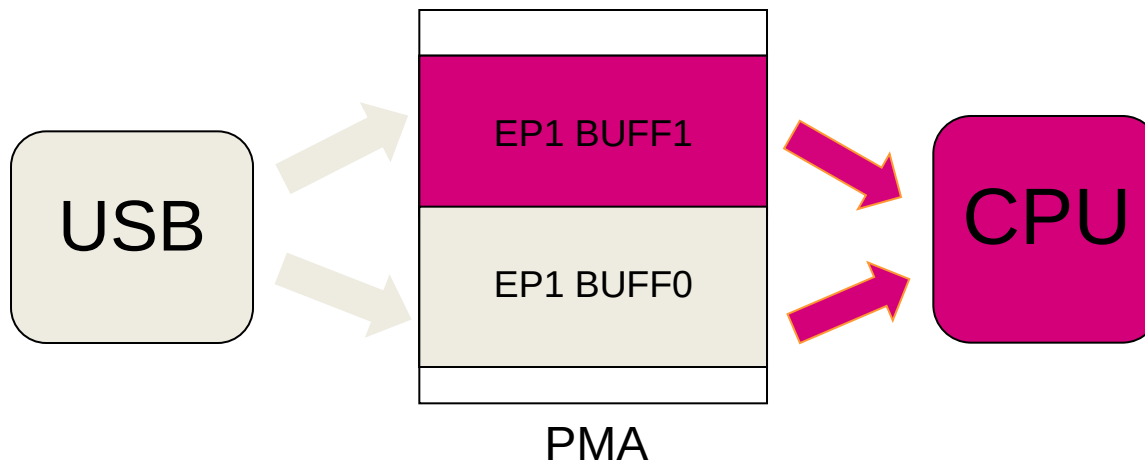
Packet Memory Area

# USB 2.0 Full-speed Device Controller

## Double-Buffering mechanism

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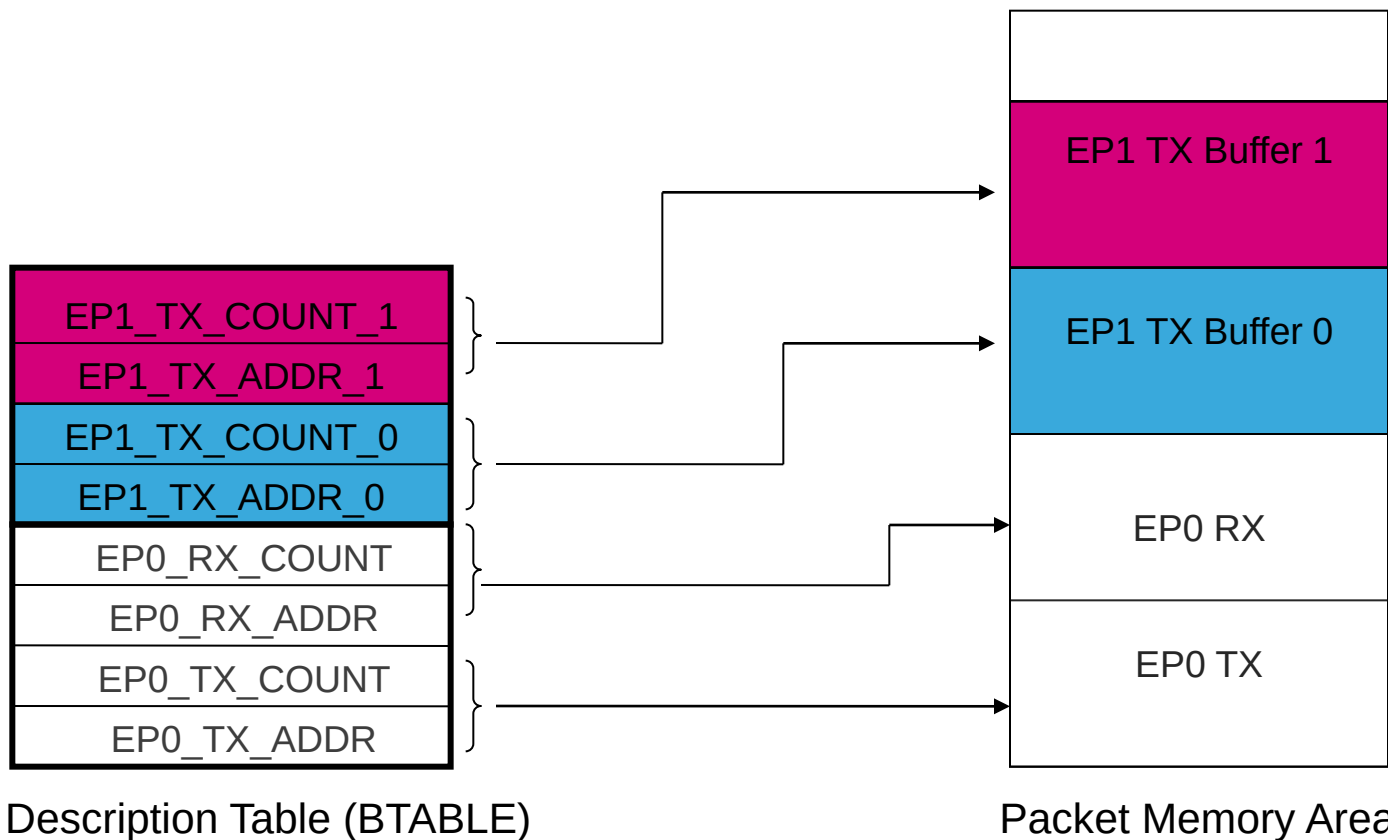
- Double buffering is used to improve the transfer performance for isochronous and bulk endpoints (in one direction only)
- Consists of using two buffers in PMA (buffer0 and buffer1), at any time CPU should be accessing one buffer (for R/W) while USB IP is accessing the other buffer
- USB swapping between buffer0 and buffer1 is done by hardware
- In double-buffered bulk transfer, If application (CPU) is too slow to give its buffer to USB, then NAK will be sent to host



# USB 2.0 Full-speed Device Controller

## Packet Memory Area with double-buffering

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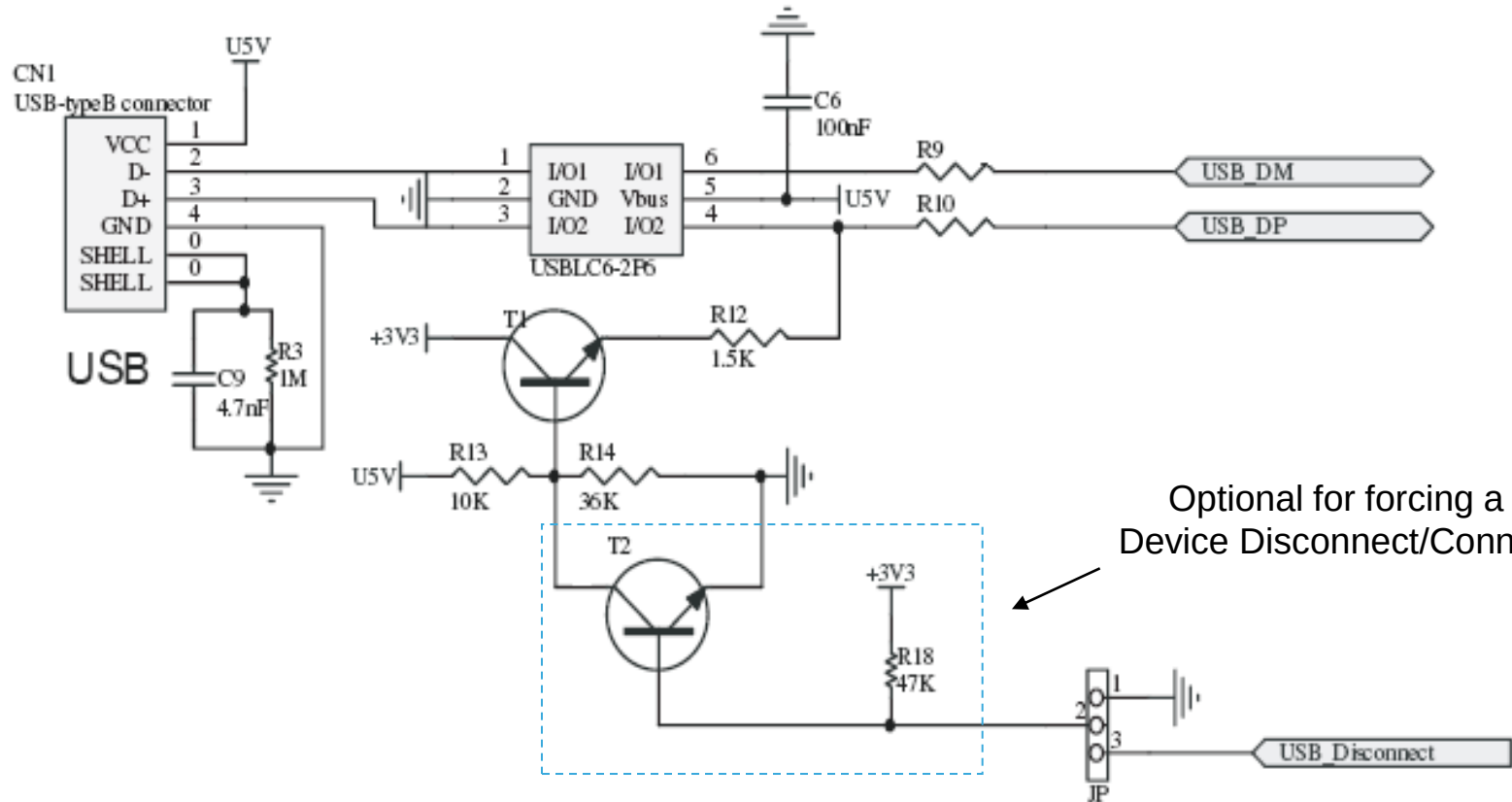
# USB 2.0 Full-speed Device Controller Suspend/Resume Interrupt

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- When no SOF is detected for 3 ms, a suspend interrupt is generated
- In the interrupt handler of the suspend interrupt, if bus powered device, the MCU should enter in low power mode in order to lower its power consumption
- In order to achieve the best low power consumption, the STM32 can enter in STOP mode (all peripherals and CPU clocks OFF)
- A host resume/reset signaling detection can wakeup the MCU from STOP mode

# USB 2.0 Full-speed Device Controller External Hardware

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# USB Library Footprints

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| Demo                                         | Flash Code + Const | RAM usage  |
|----------------------------------------------|--------------------|------------|
| Joystick demo (HID)                          | 7K Bytes           | 1400 Bytes |
| Mass Storage (Bulk)                          | 10K Bytes          | 2100 Bytes |
| CDC -Virtual Com Port<br>( Bulk+ Interrupts) | 7K Bytes           | 3400 Bytes |



Thank you !

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